

Basic Broadcast Technology — Technical Framing & Multi-Zone Design Matrix

Unit 4: Camera Operations, Composition Rules & Daktronics Layout Architecture (CTE-CTB.03.01 & CTB.03.02)

Student:

Date:

Period:

TECHNICAL DOMAIN	CAMERA & SYSTEM SETTINGS	COMPOSITION & FRAMING RULES	DAKTRONICS & ON-AIR APPLICATION
Rule of Thirds & Grid Alignment <i>Spatial Balance</i>	SYSTEM 3x3 viewfinder grid display, focal point cross-intersections.	FRAMING Align talent eye-line along top horizontal third line. Position key visual anchors off-center.	ON-AIR Prevents talent from being covered by lower-third score tickers or side stat graphics.
Headroom & Noseroom / Lead Space <i>Subject Placement</i>	SYSTEM Pan/tilt fluid drag calibration, focal headroom clearance.	FRAMING Leave 1–2 inches above head. Provide lead space in direction of subject motion/gaze.	ON-AIR Keeps anchor heads below top stadium video banners; prevents awkward graphic collisions.
Depth of Field (DoF) & Aperture <i>Optical Isolation</i>	SYSTEM Wide aperture (f/2.8 - f/4.0), telephoto focal length compression.	FRAMING Isolate main subject on sharp focal plane while softly blurring background distraction.	ON-AIR Separates talent from busy crowd/stadium noise; creates high-contrast visual focus on screen.
Multi-Zone Layout Architecture <i>Canvas Control</i>	SYSTEM Daktronics Show Control canvas splits, 16:9 native aspect ratio.	FRAMING Divide canvas into Zone 1 (Live Video), Zone 2 (Scoreboard), Zone 3 (Ad Ticker).	ON-AIR Prevents live video feed distortion or stretching across non-standard stadium LED displays.
Lower-Third & Safe Area Overlays <i>Graphic Integration</i>	SYSTEM Title safe / Action safe margins, graphic alpha transparency.	FRAMING Reserve bottom 20–25% of camera frame for graphic overlays and lower-third titles.	ON-AIR Ensures real-time scores, player stats, and sponsor bugs do not obscure anchor faces.
Motion & Exposure Calibration <i>Signal Quality</i>	SYSTEM Shutter speed (1/120s @ 60fps), manual focus assist / peaking.	FRAMING Maintain shutter speed at double frame rate to lock motion without strobing.	ON-AIR Crisp rendering on stadium LED panels; prevents motion blur on fast action replays.